SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 – SOFTWARE TESTING

EXPERIMENT 7

FOOD DELIVERY APPLICATION – NEGATIVE TEST SCENARIOS AND TEST CASES

Aim:

To write a case study to identify and analyze the requirements of Food Delivery Applications, and prepare negative test scenarios as well as test cases for the application.

1. Problem Description

A food delivery application allows customers to discover restaurants, view menus, add food items to a cart, select an address, make payment and track delivery. Restaurant availability, item availability and delivery coverage must be considered.

The Food Delivery Application is expected to provide reliable, secure and user-friendly services. The testing team verifies the stated requirements using valid and invalid inputs, records actual results, and reports any deviation from the expected behaviour.

2. Functional Requirements

1. Customers must register and log in using valid credentials.
2. The application must list restaurants available for the selected location.
3. Customers must view menu items, prices and availability.
4. Only available menu items can be added to the cart.
5. Quantity must be a positive value and respect item limits.
6. The application must calculate subtotal, taxes, delivery fee and total correctly.
7. The delivery address must fall within the restaurant's service area.
8. Orders must be confirmed only after successful payment and restaurant acceptance.

4. Negative Test Scenarios and Test Cases

Negative testing checks whether invalid, incomplete, unauthorized or boundary inputs are safely rejected with a clear message.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
NT01	Login with invalid OTP	Enter an incorrect or expired OTP	Authentication is denied.	Pass
NT02	Search with unsupported location	Enter a location outside service coverage	No deliverable restaurants are shown.	Pass
NT03	Add unavailable food	Select an item marked unavailable	Item cannot be added.	Pass
NT04	Enter zero item quantity	Set quantity to 0	Quantity is rejected.	Pass
NT05	Apply expired coupon	Enter an expired coupon code	Coupon is rejected with a reason.	Pass
NT06	Use invalid address	Enter incomplete or malformed address	Address validation message is displayed.	Pass
NT07	Place order below minimum	Cart total is below restaurant minimum	Checkout is blocked with minimum-order message.	Pass
NT08	Payment failure	Simulate declined or timed-out payment	Order is not confirmed; payment status is handled safely.	Pass

5. Detailed Execution of a Failed Test Case

Test Case ID: NT05

Precondition: The application is running in the test environment and the relevant test data is available.

Test Steps: 1. Sign in if required. 2. Open the relevant feature. 3. Perform the action: Enter an expired coupon code. 4. Observe the application response.

Expected Result: Coupon is rejected with a reason.

Actual Result: For demonstration, assume the application accepts the invalid input or does not display the required validation message.

Conclusion: The test is marked as failed for defect-report demonstration. This is an illustrative result, not a claim about a deployed application.

6. Sample Bug Report

Field	Description
Bug ID	BR-007
Linked Test Case	NT05
Severity	High (illustrative)
Priority	High (illustrative)
Bug Title	Food Delivery Application does not correctly validate the condition: Apply expired coupon.
Environment	Test environment; browser and operating system should be recorded during actual execution.
Preconditions	The application is running in the test environment and the relevant test data is available.
Steps to Reproduce	1. Sign in if required. 2. Open the relevant feature. 3. Perform the action: Enter an expired coupon code. 4. Observe the application response.
Expected Result	Coupon is rejected with a reason.
Actual Result	For demonstration, assume the application accepts the invalid input or does not display the required validation message.
Impact	Invalid data or an unauthorized operation may be accepted, resulting in incorrect processing, data integrity issues or a poor user experience.
Suggested Fix	Apply validation at both client and server sides, return a clear error message, and add regression tests for the boundary and invalid condition.
Status	Open – illustrative sample.

7. Bug Life Cycle

New → Assigned → In Progress → Fixed → Retesting → Closed. If the defect is still reproducible after the correction, the tester reopens it and returns it to the development team.

Conclusion

The test cases prepared for the Food Delivery Application provide coverage of its key functional requirements. Executing them helps identify functional gaps, improve error handling and increase confidence in the quality of the application.